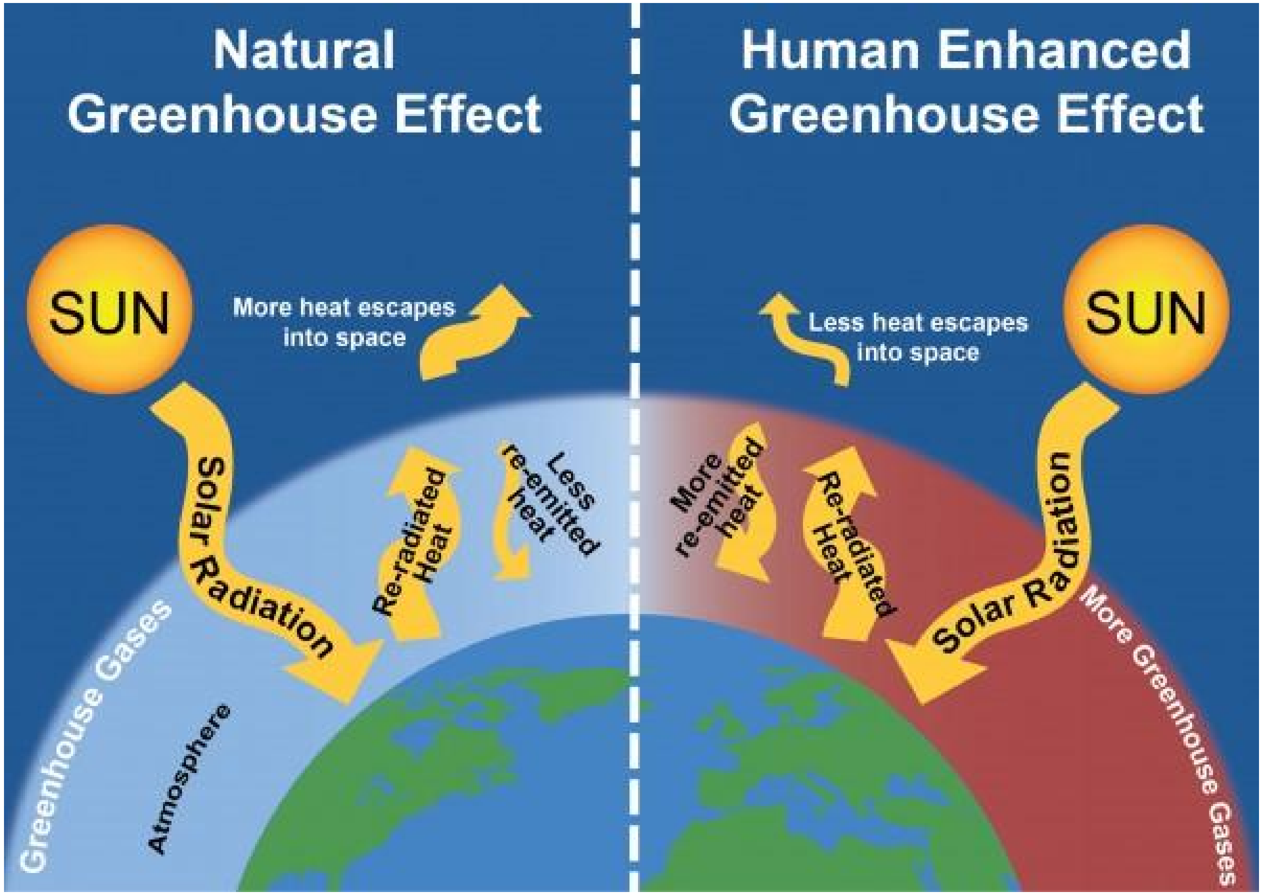
Climate Change

Evidence causes and effects

Greenhouse effect

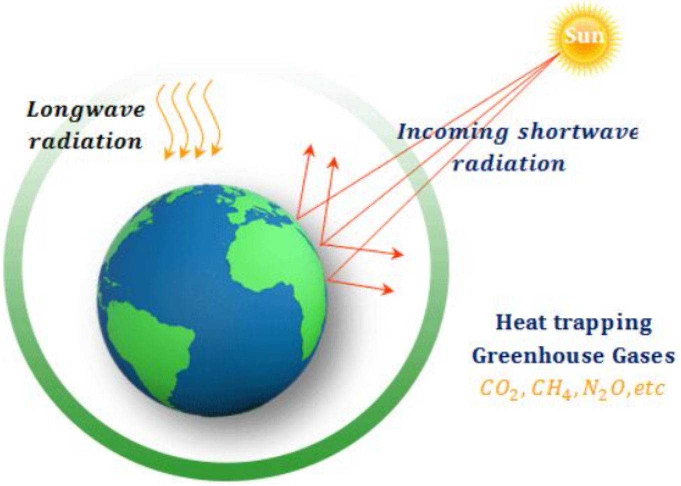
- The greenhouse effect is the natural warming of the earth that results when gases in the atmosphere trap heat from the sun that would otherwise escape into space.
- Without the natural greenhouse effect, the heat emitted by the Earth would simply pass outwards from the Earth's surface into space

Greenhouse effect



Greenhouse Gases

- A greenhouse gas is called that because it absorbs infrared radiation from the Sun in the form of heat, which is circulated in the atmosphere and eventually lost to space.



Greenhouse Gases

- **Water vapour (H_2O)**
- **Carbon dioxide (CO_2)** – Carbon dioxide enters the atmosphere through burning fossil fuels (coal, natural gas, and oil), solid waste, trees and other biological materials, and also as a result of certain chemical reactions (e.g., manufacture of cement). Carbon dioxide is removed from the atmosphere (or "sequestered") when it is absorbed by plants as part of the biological carbon cycle.
- **Nitrous oxide(N_2O)** — Nitrous oxide is emitted during agricultural, land use, industrial activities, combustion of fossil fuels and solid waste, as well as during treatment of wastewater.

Greenhouse Gases

- **Methane (CH_4)** – Methane is emitted during the production and transport of coal, natural gas, and oil. Methane emissions also result from livestock and other agricultural practices, land use and by the decay of organic waste in municipal solid waste landfills.

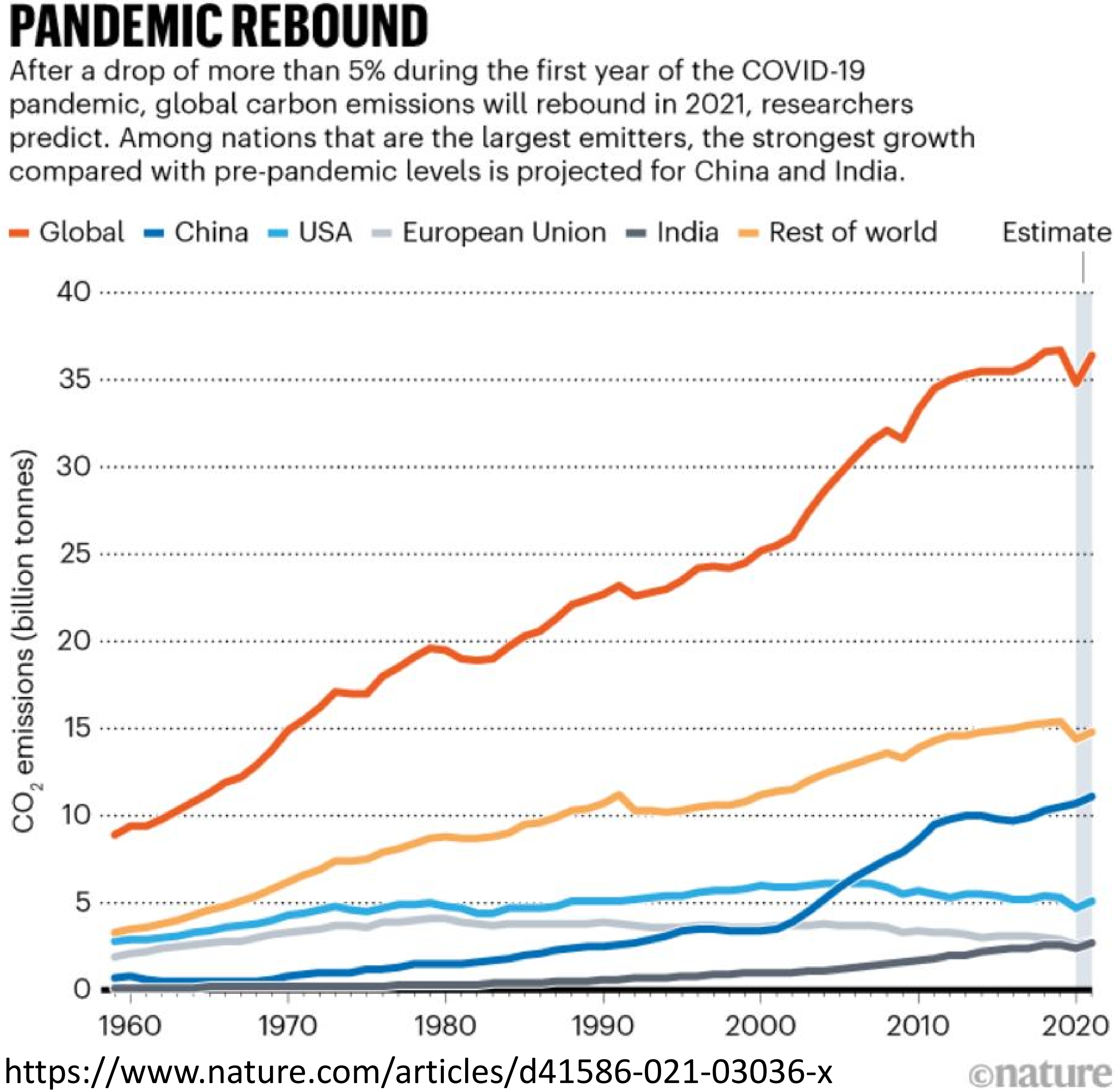
- **Fluorinated gases:** Hydrofluorocarbons, perfluorocarbons, sulfur hexafluoride, and nitrogen trifluoride are synthetic, powerful greenhouse gases that are emitted from a variety of industrial processes. Fluorinated gases are sometimes used as substitutes for stratospheric ozone-depleting substances (e.g., chlorofluorocarbons, hydrochlorofluorocarbons, and halons). These gases are typically emitted in smaller quantities, but because they are potent greenhouse gases, they are sometimes referred to as High Global Warming Potential gases ("High GWP gases").

Greenhouse Gases

Global warming potential or GWP: the amount of heat these gases can absorb and re-radiate

According to GWP CH_4 is 23 times more effective and N_2O is 296 times more effective than CO_2.

A **carbon dioxide equivalent** or **CO_2 equivalent**, abbreviated as **CO_2-eq** is a metric measure used to compare the emissions from various greenhouse gases on the basis of their global-warming potential (GWP), by converting amounts of other gases to the equivalent amount of carbon dioxide with the same global warming potential



https://www.nature.com/articles/d41586-021-03036-x

Global Warming

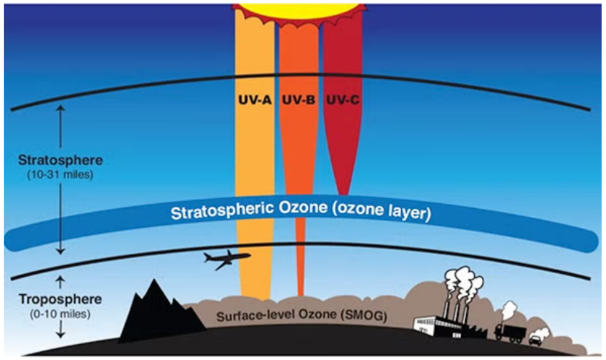
- Global warming is the long-term heating of Earth's climate system observed since the pre-industrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere.
- Human activities are estimated to have increased Earth's global average temperature by about 1 degree, a number that is currently increasing by 0.2 degrees Celsius per decade.

Climate Change

- Climate change is a long-term change in the average weather patterns that have come to define Earth’s local, regional and global climates.
- Changes observed in Earth’s climate since the early 20th century are primarily driven by human activities
- Natural processes can also contribute to climate change,

Climate Change

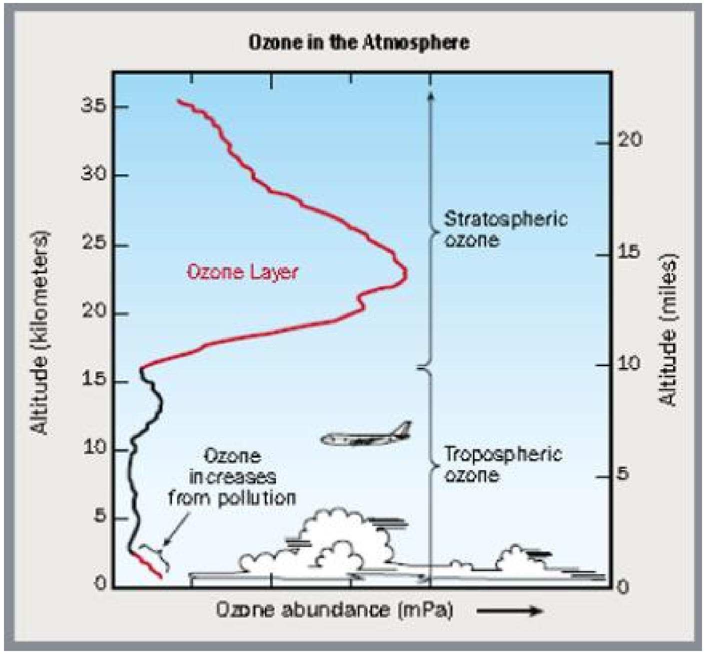
- Climate change in IPCC usage refers to any change in climate over time, whether due to natural variability or as a result of human activity.
- UN Framework Convention on Climate Change, climate change refers to a change of climate that is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and that is in addition to natural climate variability observed over comparable time periods.



Ozone layer

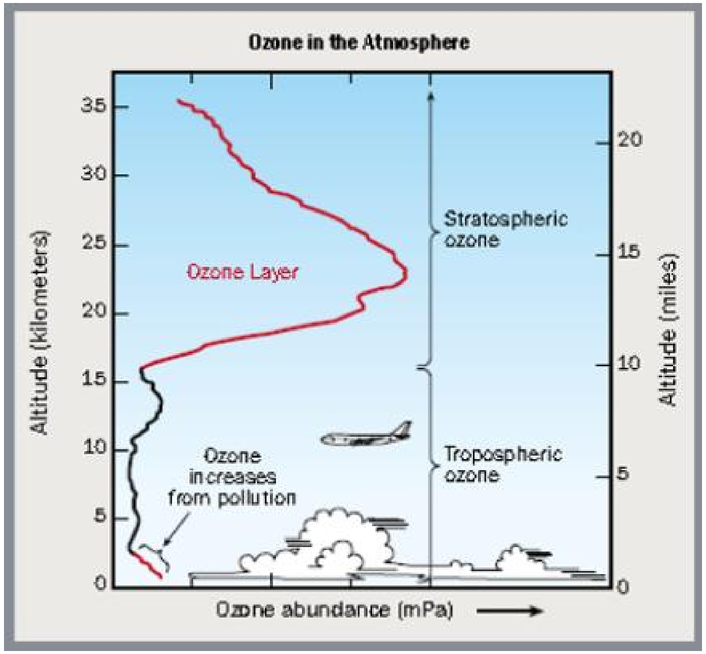
- The ozone layer in the stratosphere absorbs UVB portion of the radiation from the sun, preventing it from reaching the Earth's surface.
- UVB has been linked to many harmful effects, including skin cancers, cataracts, and harm to some crops and marine life.

Ozone layer depletion



- Ozone concentrations in the atmosphere vary naturally with sunspots, seasons, and latitude.
- Each natural reduction in ozone levels has been followed by a recovery.
- Beginning in the 1970s, however, scientific evidence showed that the ozone shield was being depleted well beyond natural processes.

Ozone layer depletion



- When chlorine and bromine atoms come into contact with ozone in the stratosphere, they destroy ozone molecules.
- One chlorine atom can destroy over 100,000 ozone molecules before it is removed from the stratosphere. Ozone can be destroyed more quickly than it is naturally created.

Ozone depleting substances

- ODS releasing chlorine
  - Chlorofluorocarbons (CFC's)
  - Hydrochlorofluorocarbons (HCFC's)
  - Carbon tetrachloride
  - Methyl chloroform
- ODS releasing Bromine
  - Bromofluorocarbons (Halons)
  - methylbromide

What is Climate Change?

- The change in our climate and weather systems being caused by the warming of the earth

- Today the earth is hotter than it has been in 2,000 years

  - 1990s was the warmest decade
  - 1998 was the warmest year
  - Snow cover has reduced by 10% in the last 40 years

- Climate Change is a global issue – it affects the whole planet

Greenhouse Gases (GHGs)

1. Carbon dioxide (CO_2)
2. Methane (CH_4)
3. Nitrous oxide (NOx)
4. Hydroflourocarbons (HFCs)
5. Perflourocarbons (PFCs)
6. Sulfur hexaflouride (SF_6)

Cause of Global Warming

Release of GHGs into the atmosphere

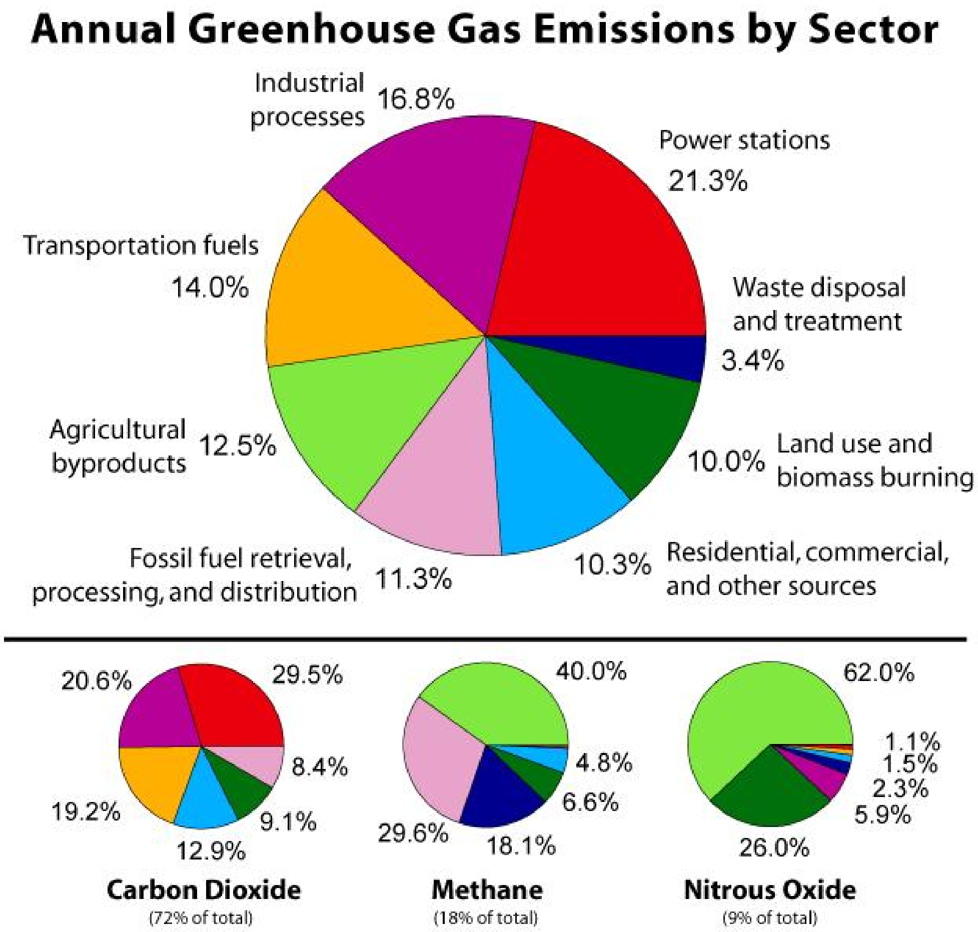
Natural

- Release of methane (CH_4) from arctic tundra and wetlands

Causes of Global Warming

Anthropogenic

- Use of fossil fuels (industry, transportation)
- Land use change (agriculture, deforestation)



Global GHG emissions by sector for year 2000
(Source: http://ghg.unfccc.int/index.html)

How do we know the climate is changing?

- Intergovernmental Panel on Climate Change (IPCC) established by WMO and UNEP in 1988 – it is the science authority for UNFCCC
- Objective – to assess the scientific, technical and socio-economic information for understanding the scientific basis of risk of human-induced climate change, its potential impacts and options for adaptation and mitigation
- IPCC does not carry out research nor monitor climate related data or other relevant parameters. It bases its assessment on peer reviewed and published scientific/technical literature
- Assessment Reports, Working Groups

IPCC Secretariat is hosted by WMO in Geneva, Switzerland

IPCC 4th Assessment Report, 2007

"Warming of the climate system is unequivocal, as is now evident from observations of increases in global average air and ocean temperatures, widespread melting of snow and ice, and rising global average sea level"

- **Eleven of the last twelve years (1995-2006) rank among the 12 warmest years in the instrumental record of global surface temperature (since 1850)**
- **Updated 100-year linear trend of 0.74°C for 1906-2005**

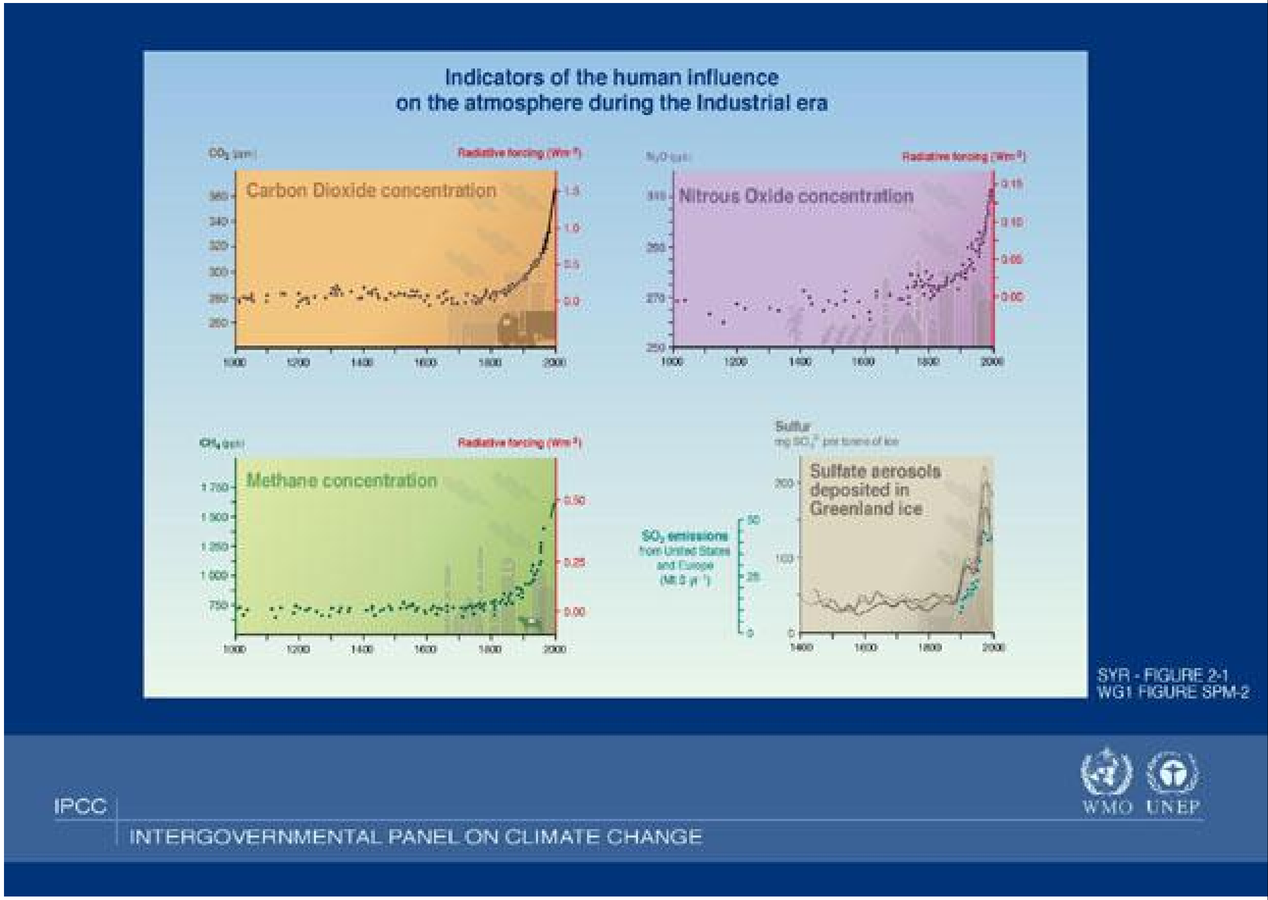
Evidence of changes in Earth's climate

IPCC 4th Assessment Report, 2007

1. Atmospheric concentration of CO_2

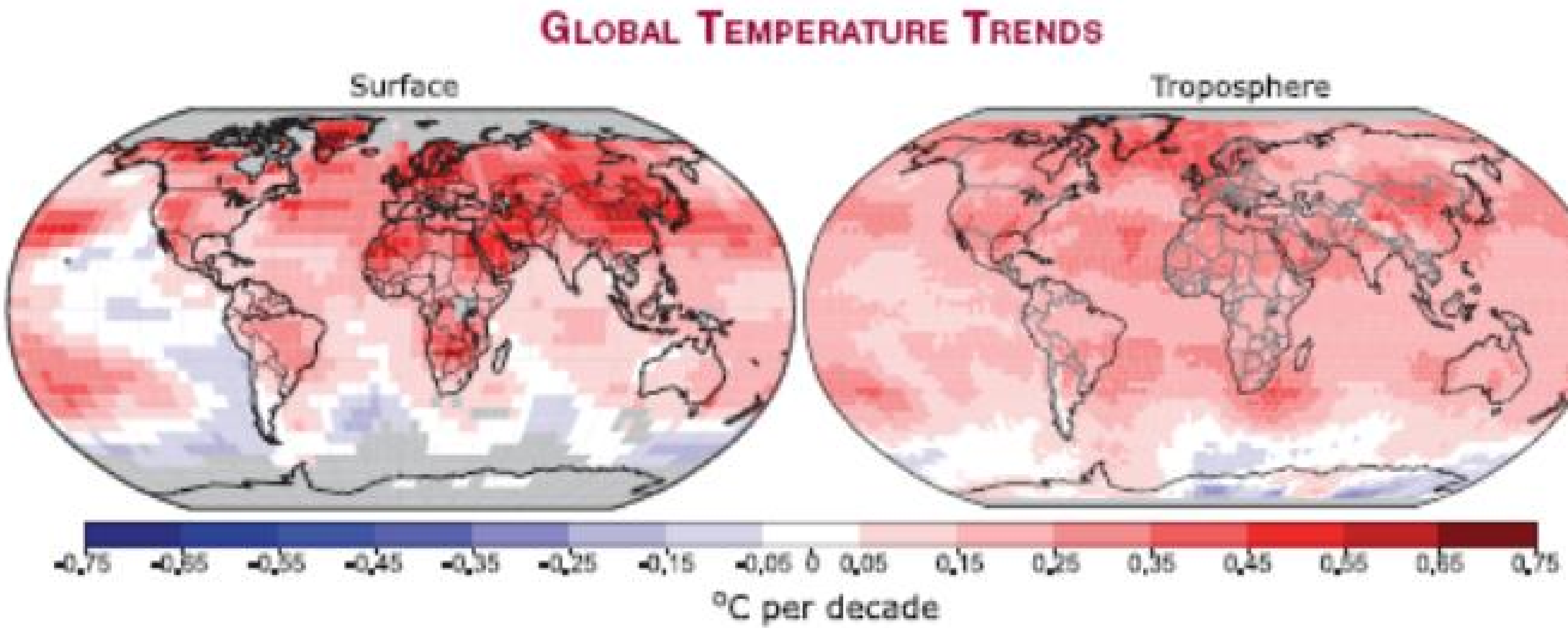
- 280 ppm for the period 1000 –1750
- 379 ppm in year 2000 (368 ppm reported in IPCC TAR)

Annual CO_2 concentration growth rate was larger during the last 10 years (1995-2005 average: 1.9 ppm per year), than it has been since the beginning of continuous direct atmospheric measurements (1960-2005 average: 1.4 ppm per year)



2. Global mean surface temperature

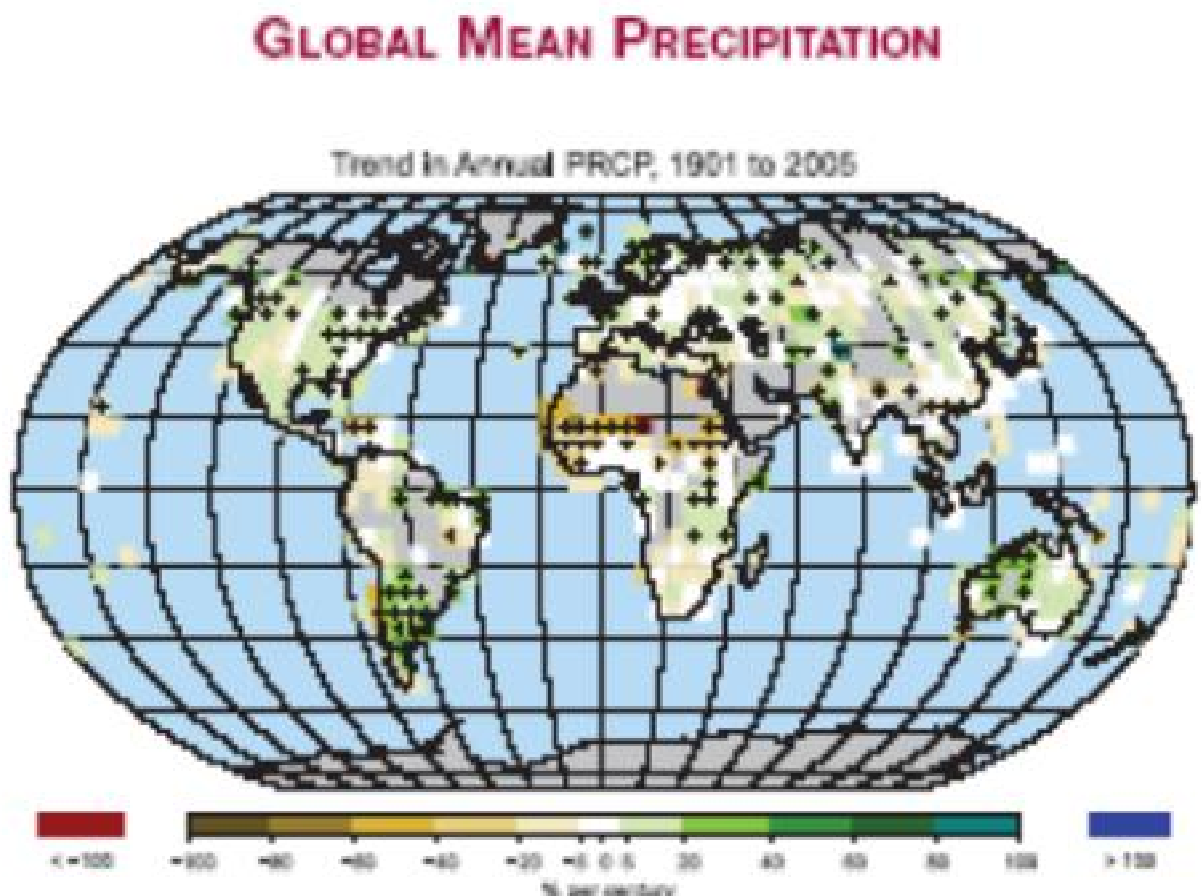
- 0.74°C increase over the 20th century (land areas warmed more than the oceans)



Patterns of linear global temperature trends over the period <u>1979 to 2005</u> estimated at the surface (left), and for the troposphere from satellite records (right). (IPCC 4th Assessment Report)

3. Continental precipitation

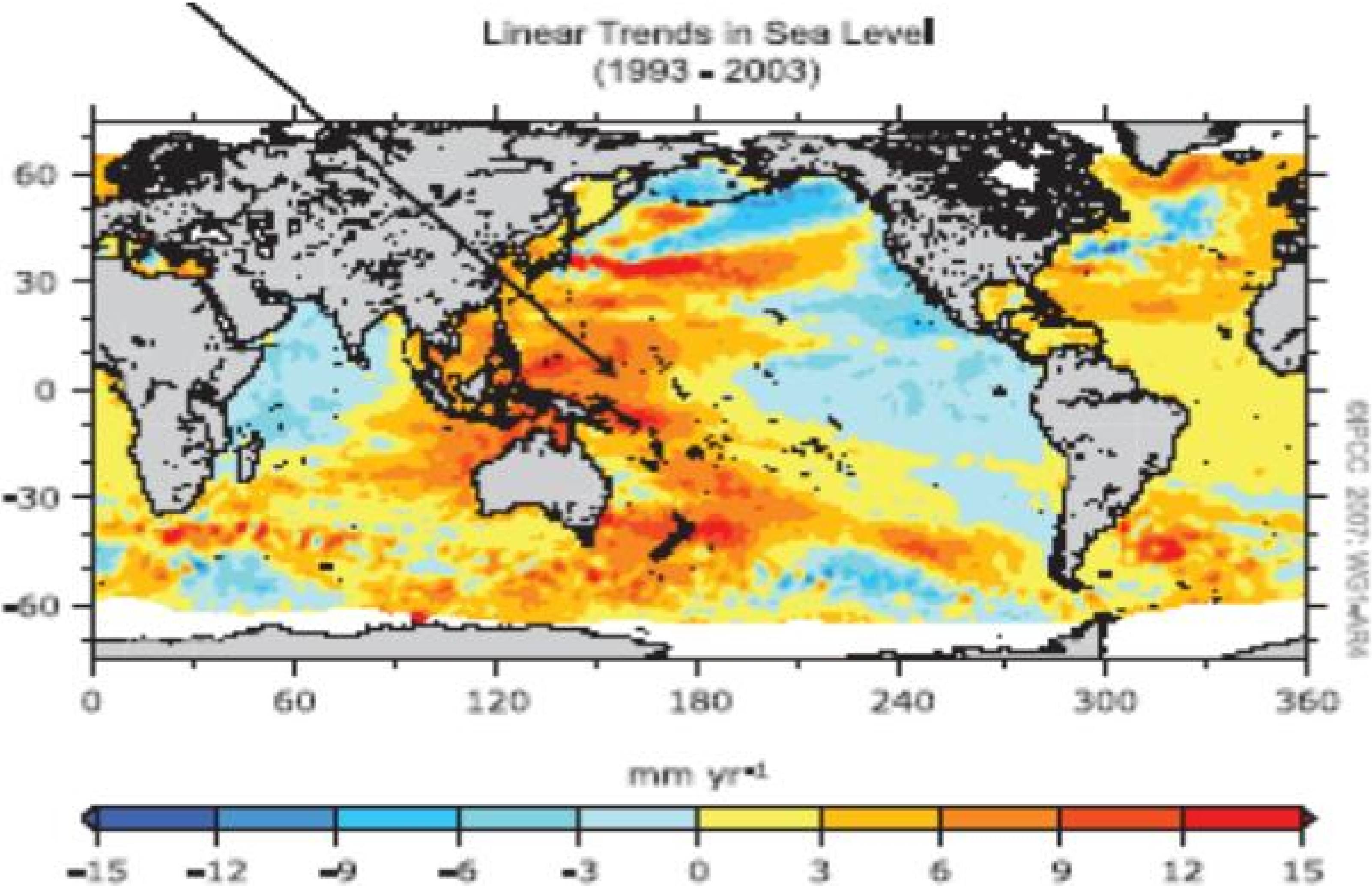
- Significant increase in North & South America, northern Europe, north & central Asia
- Drying in Sahel, Mediterranean, southern Africa, parts of South Asia



Distribution of linear trends of annual land precipitation amounts over the period 1901 to 2005 (% per century) (IPCC 4th Assessment Report)

4. Global sea level rise

Average rate of 1.8 mm per year (1961-2003)

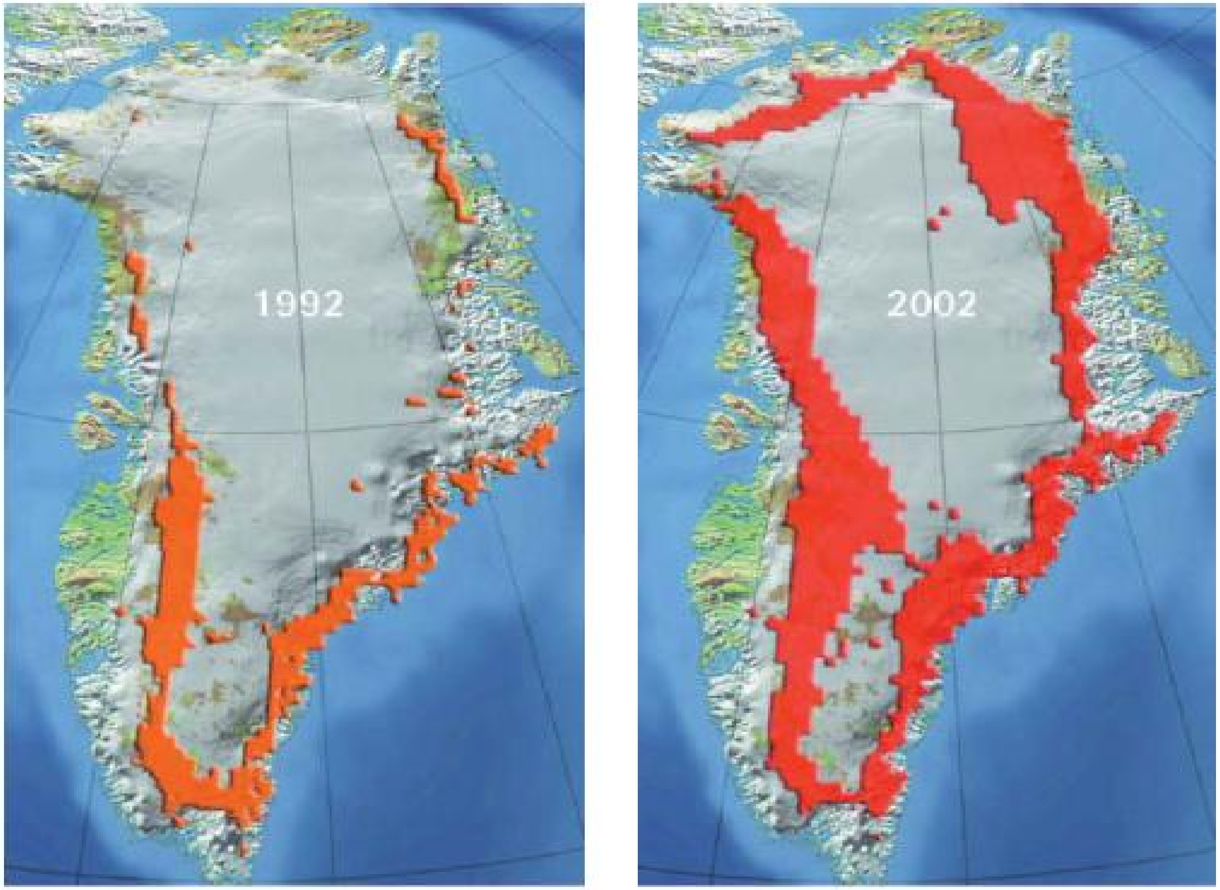


5. Arctic sea-ice extent and thickness

Decrease in extent – 2.7% per decade since 1978

Max. area of seasonally frozen ground decreased by 7% in Northern Hemisphere since 1900

Figure 2.20 Seasonal melting of the Greenland Ice Sheet

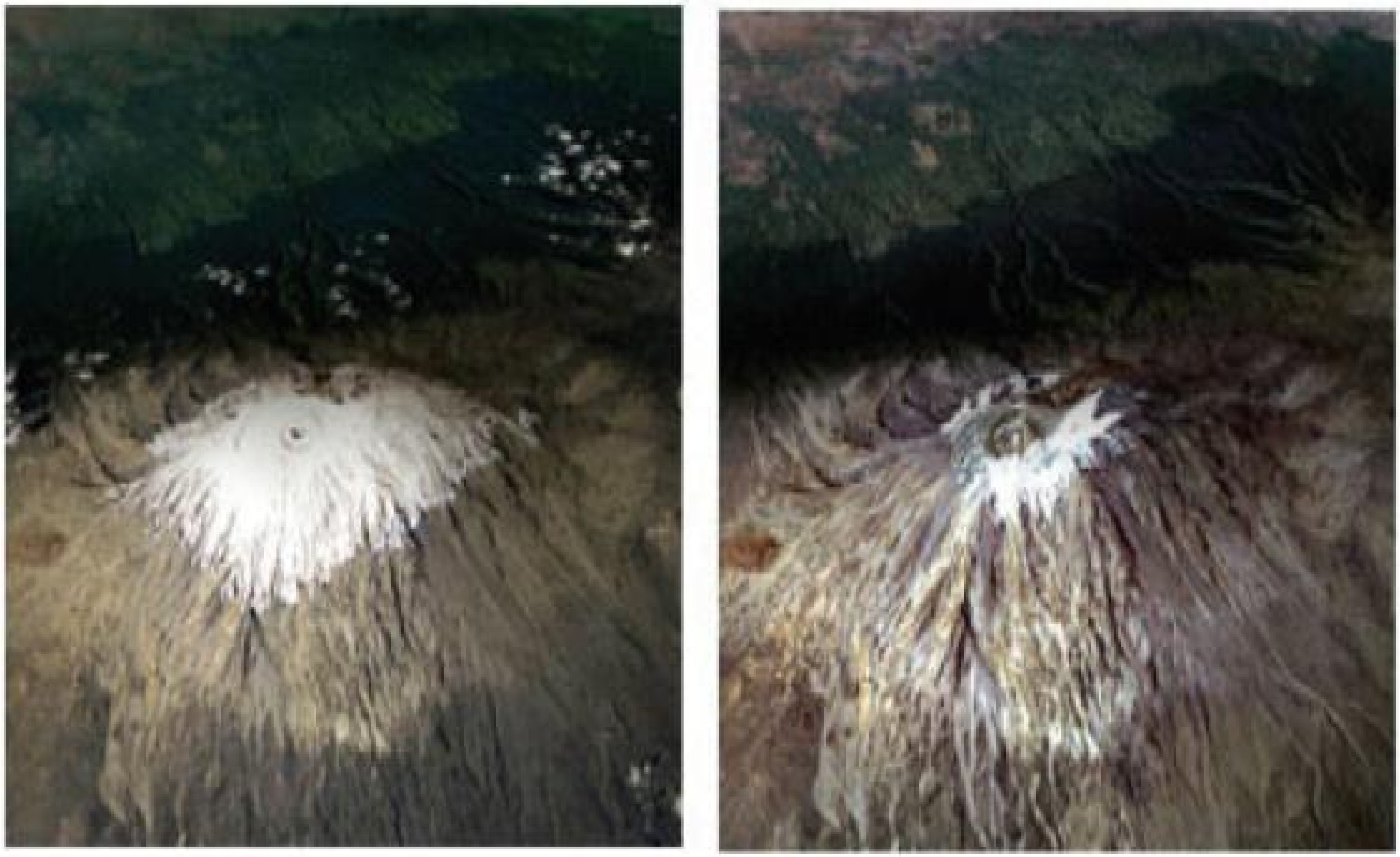


Note: The areas in orange/red are the areas where there is seasonal melting at the surface of the ice sheet.

Source: Steffen and Huff 2005

6. Non-polar glaciers

Widespread retreat during the 20th century



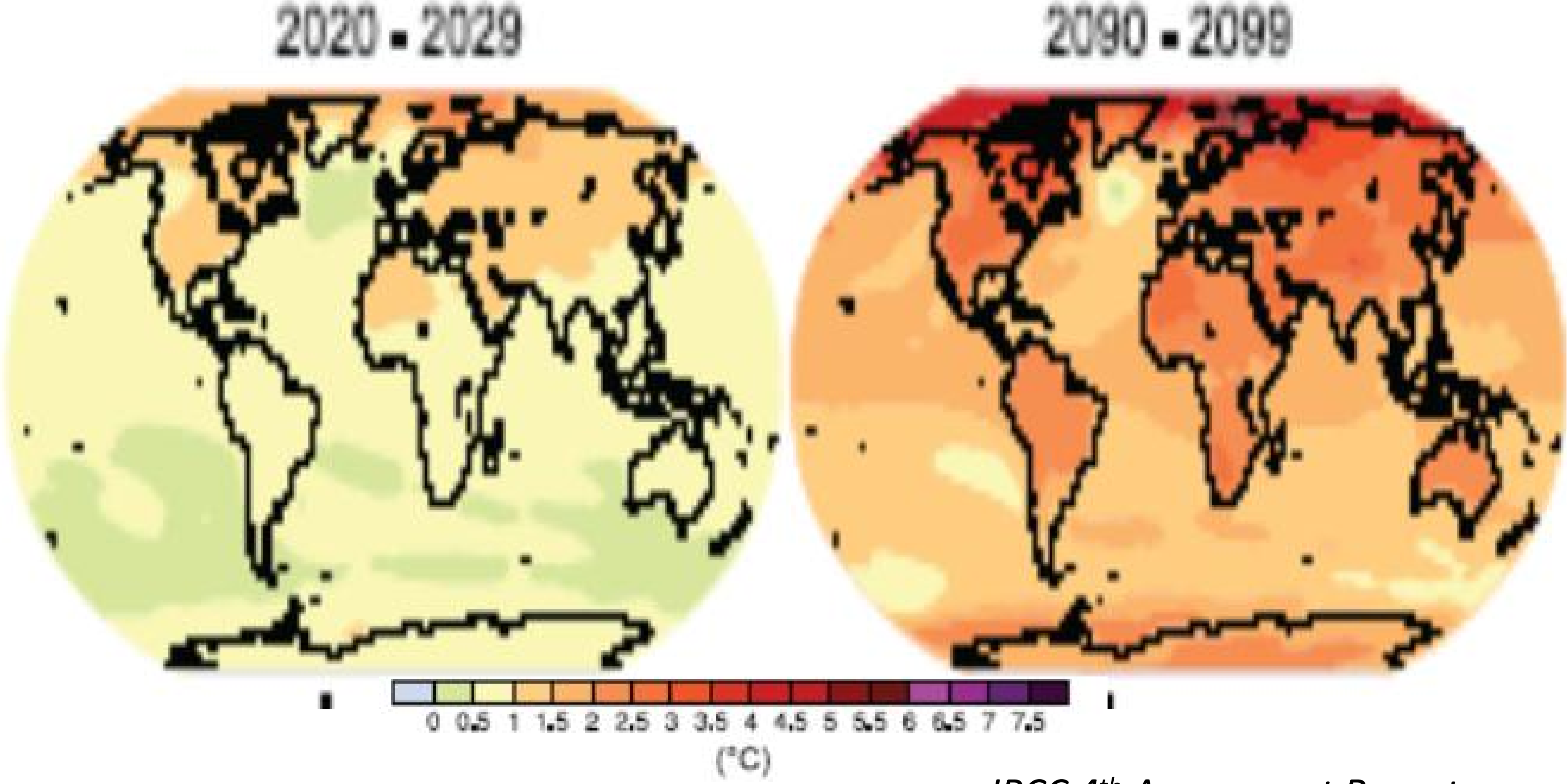
Mt. Kilimanjaro-- February 17, 1993 Mt. Kilimanjaro-- February 21, 2000

Projections of Future Changes in Climate

- For the next 2 decades, a warming of about 0.2^{o}C per decade is projected
- Increases in amount of precipitation are very likely in high latitudes, while decreases are likely in most subtropical land regions
- Hot extremes, heat waves and heavy precipitation events are expected to become more frequent
- Likely that future tropical cyclones will become more intense
- Snow cover is projected to contract
- Past and future anthropogenic CO_2 emissions will continue to contribute to warming and SLR for more than a millennium, due to time scales required for removal of this gas from the atmosphere

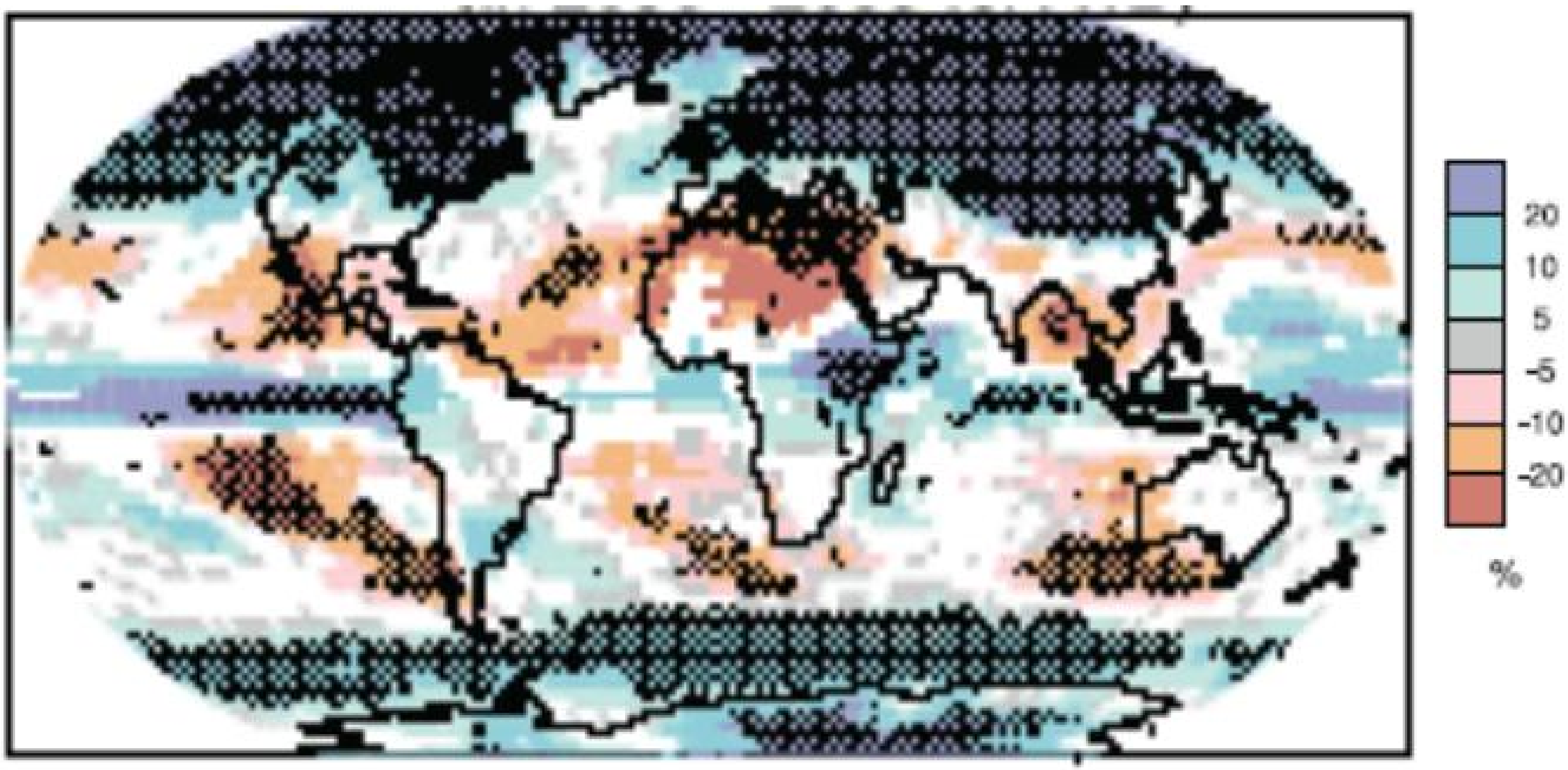
IPCC 4th Assessment Report

PROJECTIONS OF SURFACE TEMPERATURES



IPCC 4th Assessment Report

Precipitation – Projected Change in 2090 - 2099



(darkened areas indicate projections for which more than 90% of the models agree on the sign of the change)   IPCC 4th AR

Impacts

Water resources

Climate change will exacerbate water shortages in many water-scarce areas of the world

- **Demand for water is increasing due to population growth and economic development**
- **Substantial reduction of available water in many of the water-scarce areas of the world, but increase in some other areas**
- **Freshwater quality would generally be degraded by higher water temperatures, but this may be offset in some regions by increased flows**

Agriculture

- Changes in temperatures and precipitation patterns will lead to changes in crop yields
- Length of growing season will change
- Droughts, extreme events will decrease yields
- Biodiversity shifts – changes in crop varieties
- New pests and diseases
- Food security

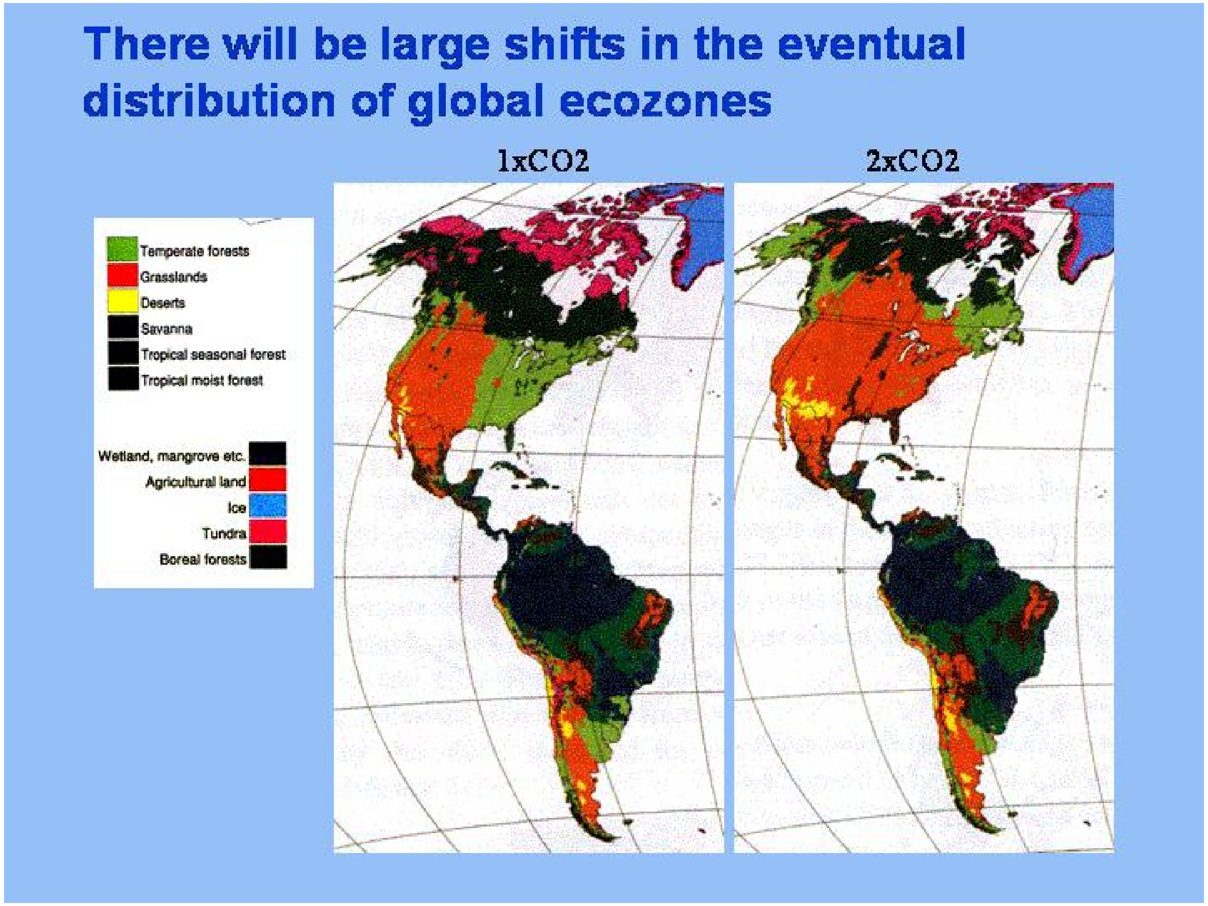
Health

- Thermal stress – heat waves, cold spells
- Extreme events, weather disasters (personal injuries, damage & disruption to infrastructure)
- Infectious diseases (malaria, dengue)
- Air pollution - impact of some air pollutants (SO_2, CO, NO_2, O_3) on health is more evident during the summer or high temperatures

Ecological systems

Biological systems have already been affected on the regional scale

Structure and functioning of ecological systems will be altered & biodiversity will decrease especially in niche systems (e.g. alpine, arctic)



Coastal areas

- Most sensitive coasts - Beaches, salt marshes, mangrove swamps, deltas, coral reefs, lagoons

Sea Level Rise

- Likely sea level rise during the 21st century - 5 mm per year
- Low lying areas inundated, small island states may disappear, salt water intrusion into aquifers, floods
- Low-lying coastal cities - Shanghai, Jakarta, Tokyo, Manila, Bangkok, Karachi, Mumbai, and Dhaka will be at the forefront of impacts

Species

Plant and animal ranges will shift poleward and up in elevation

Vulnerable species will be increasingly threatened by changing habitat and food supply

Extreme events

- Hurricanes and storms - increase in tropical cyclone peak wind intensities, mean and peak precipitation intensities
- Floods (Pakistan floods 2005)
- Droughts (Pakistan drought 1999-2001)
- Increased frequency of GLOFs and landslides
- Heat and cold waves (European heat wave 2003)
- Climatic variability

Climate Change

- https://www.youtube.com/watch?v=xWrIb-9KzaA
- https://www.youtube.com/watch?v=-D_Np-3dVBQ
- https://www.youtube.com/watch?v=xWrIb-9KzaA
- https://www.un.org/en/climatechange/science/causes-effects-climate-change